# ASEEM


AI/ML Engineer | Applied AI & Intelligent Systems

Pune, Maharashtra, India | +91 7428515564 | aseemajit@gmail.com | https://www.linkedin.com/in/aseem-kumar-190b8876 | https://github.com/aseem395


## CAREER SUMMARY:


AI/ML Engineer with research-focused experience in medical image segmentation and applied AI systems. Developed and evaluated multiple U-Net architectures for diabetic retinopathy detection (under publication) and currently building machine learning models for PCOS/PCOD detection. Strong engineering background with experience delivering production-ready applications, AI-powered chatbots, and scalable backend systems.


## WORK EXPERIENCE:

**Bondigitally**

AI & Automation Developer Intern | Feb 2026 – May 2026

- Built AI-powered chatbots for customer engagement and lead generation
- Designed automation workflows integrating APIs and business tools
- Developed websites, internal tools, and lightweight business applications

**M/s eHealth-Care**

App Developer Intern | Jun 2025 – Sep 2025

- Developed full-stack web applications using FastAPI and PostgreSQL
- Built frontend interfaces and backend systems for booking and user management
- Worked on integration, testing, and debugging across application layers

**Pathnovo Solutions**

React Native Developer Intern (Sole Mobile Engineer) | Jul 2025 – Jan 2026

- Sole mobile engineer responsible for end-to-end application development and deployment
- Built production-ready applications with authentication, booking, payments, and real-time features
- Integrated REST APIs and optimized application performance and stability

## PROJECTS:

Diabetic Retinopathy Detection using U-Net Variants (Under Publication)

- Compared 8 U-Net architectures for retinal image segmentation and optical disc analysis
- Performed segmentation, masking, preprocessing, and augmentation on retinal datasets
- Evaluated models using Dice coefficient and IoU metrics
- Ongoing work includes cup and optic nerve segmentation using deep learning
- Built using Python and TensorFlow/PyTorch

PCOS / PCOD Detection using Machine Learning (Ongoing)

- Developing predictive machine learning models using structured clinical datasets
- Performing feature engineering, preprocessing, and model comparison
- Evaluating models using accuracy, precision, recall, and F1-score
- Exploring real-world deployment possibilities for early diagnosis support

Actikev – Mobile Application

- Built and deployed a production-grade React Native application
- Implemented authentication, booking flows, payments (Stripe), and user management
- Integrated REST APIs and real-time data handling
- Delivered via Expo EAS and TestFlight

## EDUCATION:

2023 - 2027 | Bharati Vidyapeeth Deemed to be University College of Engineering, Pune

Bachelor of Technology (B.Tech) – Electronics and Communication Engineering

## SKILLS:

- AI/ML & Deep Learning: Medical Image Segmentation, U-Net Architectures, Deep Learning, Model Evaluation (Dice, IoU)
- Machine Learning: Classification, Feature Engineering, Model Comparison, Performance Evaluation
- Languages & Frameworks: Python, TensorFlow / PyTorch, JavaScript
- Backend & Tools: FastAPI, REST APIs, Git, GitHub, Postman, TestFlight